European Network of
Transmission System Operators
For Electricity

High Level Concept

Energy Communication Platform
Version 4.16.0

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Energy Communication Platform, version 4.16.0

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2. Executive Summary

This document introduces the ECP and describes its key design concepts and most important features. The document is integral part of the ECP technical documentation. More detailed information about ECP can be found in

- Functional Specification, which describes ECP from the user point of view
- System Design with low-level decomposition to internal system components

Document is targeted to managers and solution architects that consider the ECP as communication platform for their solutions.

This document is targeted at Business Users.

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3. Revision History

Version	Date	Author	Description
4.0.0	24.3.2016	Jiří Dudek, Jiří Neuman	ECP 4.0 Phase 1 version
4.0.1	13.6.2016	Jiří Dudek, Jiří Neuman	Introduce Revision History
4.2.0	21.03.2018	Josef Brož	Version update
4.3.0	26.06.2018	Josef Brož	Version update
4.4.0	14.06.2019	Jakub Malý	Version update
4.5.0	22.08.2019	Jiří Kakrda	Version update
4.6.0	05.11.2019	Jiří Kakrda	Version update
4.6.1	18.03.2020	Anna Langrová	Version update
4.7.0	28.06.2020	Lukáš Voříšek	Version update
4.7.1	31.07.2020	Anna Langrová	Version update
4.7.2	16.11.2020	Lukáš Voříšek	Version update
4.8.0	14.05.2021	Lukáš Voříšek	Version update
4.8.1	09.07.2021	Lukáš Voříšek	Version update
4.8.2	03.12.2021	Jiří Kakrda	Version update
4.9.0	26.05.2022	Jiří Kakrda	Version update
4.10.0	07.11.2022	Daniel Janota	Hibernate
4.11.0	16.05.2023	Jiří Kakrda	Version update
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4.14.0	10.06.2024	Jiří Kakrda	Artemis
4.15.0	21.10.2024	Jiří Kakrda	Version update

4.16.0	06.05.2025	Jiří Kakrda	Version update

4. High Level Concept

4.1. General Overview

The purpose of Energy Communication Platform is to provide message delivery capabilities with following additional key features:

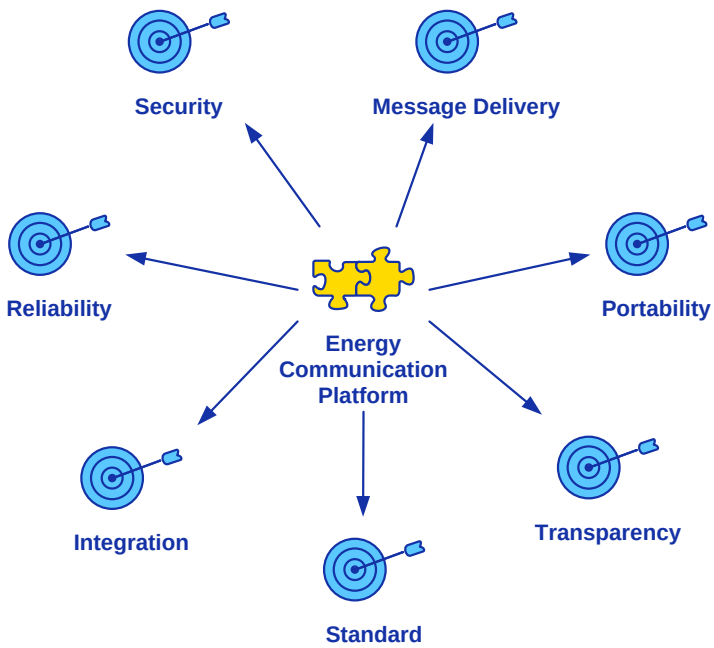

Figure 1: Key Features

Security – Only recipient of the message can read the message content. The sender of any message can be unambiguously verified.

Reliability – The message delivery is guaranteed.

Integration – The ECP functionality for sending and receiving messages can be integrated with wide variety of technologies.

Standard – ECP is the implementation of ENTSO-E MADES standard, which defines the technical aspects of communication between entities in the energy sector.

Transparency – Any message transported by ECP can be tracked down to gather trustworthy information about the state of delivery

Portability – The ECP can be installed on most of widely-used operation systems

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4.2. Main Feature: Message Delivery

Main feature of Energy Communication Platform is message delivery:

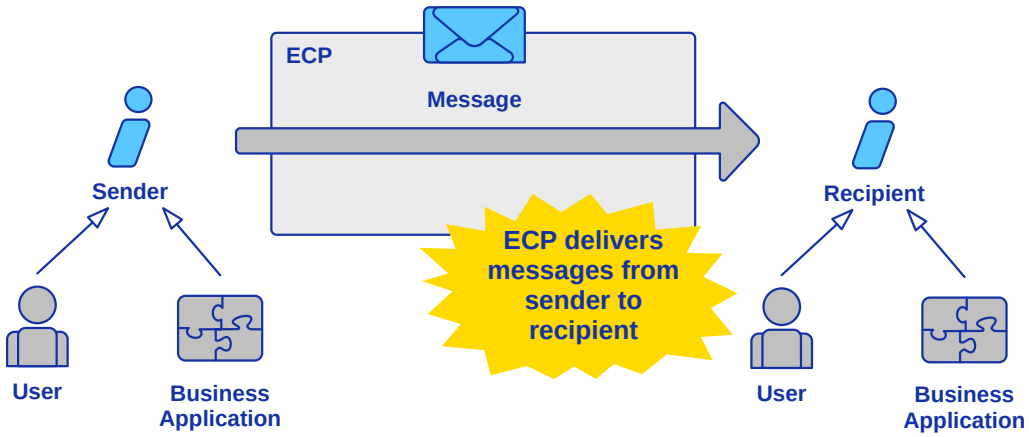

Figure 2: Message Delivery

ECP delivers messages from a sender to a recipient. Both sender and recipient can be either User or Business Application – users can connect to ECP via graphical screens, business applications via programming interface or via file system folder monitored by ECP.

The sender and recipient view the ECP only through the available interface. Any internal complications of ECP network are completely hidden from the clients.

The message transported between sender and recipient can be any text or binary data. Alongside with the message, ECP also transfers message metadata consisting of among others information about sender and recipient, code of message business type, unique message id and code of sending business application.

4.3. Architecture Concept: Four Main Components

Energy Communication Platform is internally composed of four main components – Endpoint, Internal Broker, Component Directory and Broker.

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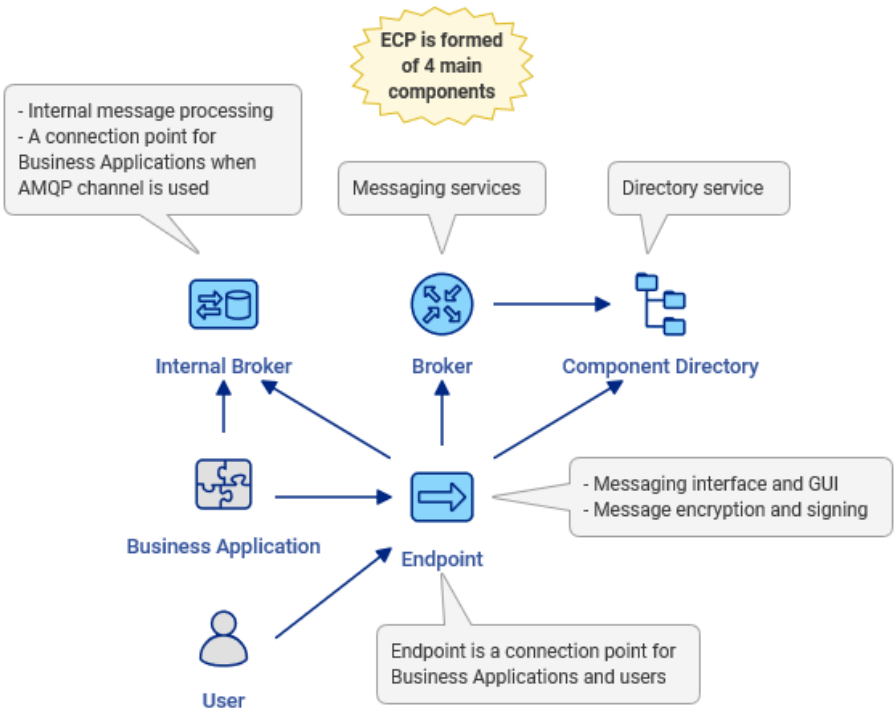

Figure 3: Main Components

From the users' point of view the crucial component is the Endpoint, which provides the user interface for sending and receiving messages and API for integration with business applications.

The Internal Broker provides messaging services for the internal message processing in Endpoint. Endpoint must be always connected to Internal Broker to be able to process messages. Multiple Endpoints can be connected to one Internal Broker instance. Business Application can also send and receive messages from Internal Broker in case the AMQP integration channel is used.

The Component Directory component serves as a central part of ECP network managing the information about all connected ECP component – Endpoints, Brokers, and other Component Directories.

The Broker component provides messaging services to Endpoints.

4.4. Additional Architecture Concepts

4.4.1. Distributed Architecture

The ECP network may consist of multiple interconnected Component Directories, each taking care of part of ECP network:

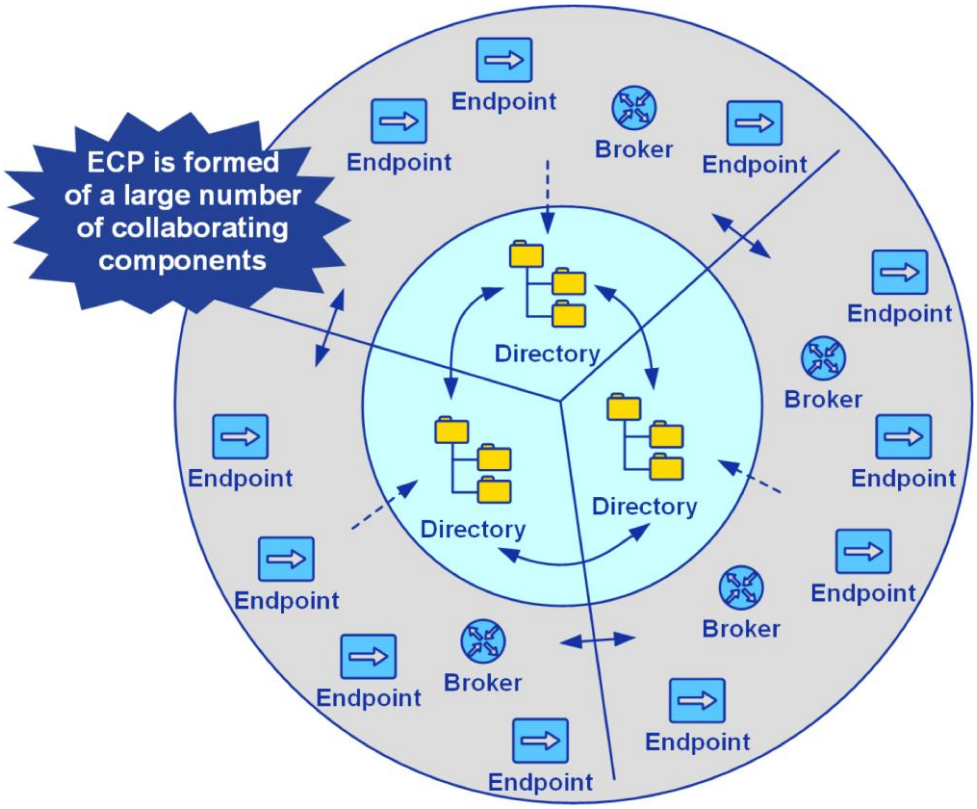

Figure 4: Distributed Architecture

ECP network itself is composed as a collaboration of large number components with the Component Directories in the centre. ECP has a distributed architecture; it does not have any single central component. All Component Directories have equal responsibilities; each manages its own part of ECP network.

The information about all connected Endpoints and Brokers is regularly shared between the Component Directories, so that Endpoints connected to different Component Directories can mutually communicate.

4.4.2. Extensibility

Available functionality of ECP Endpoints can be extended by a mechanism of plugins; this way the ECP can be adjusted to fit almost any possible integration need.

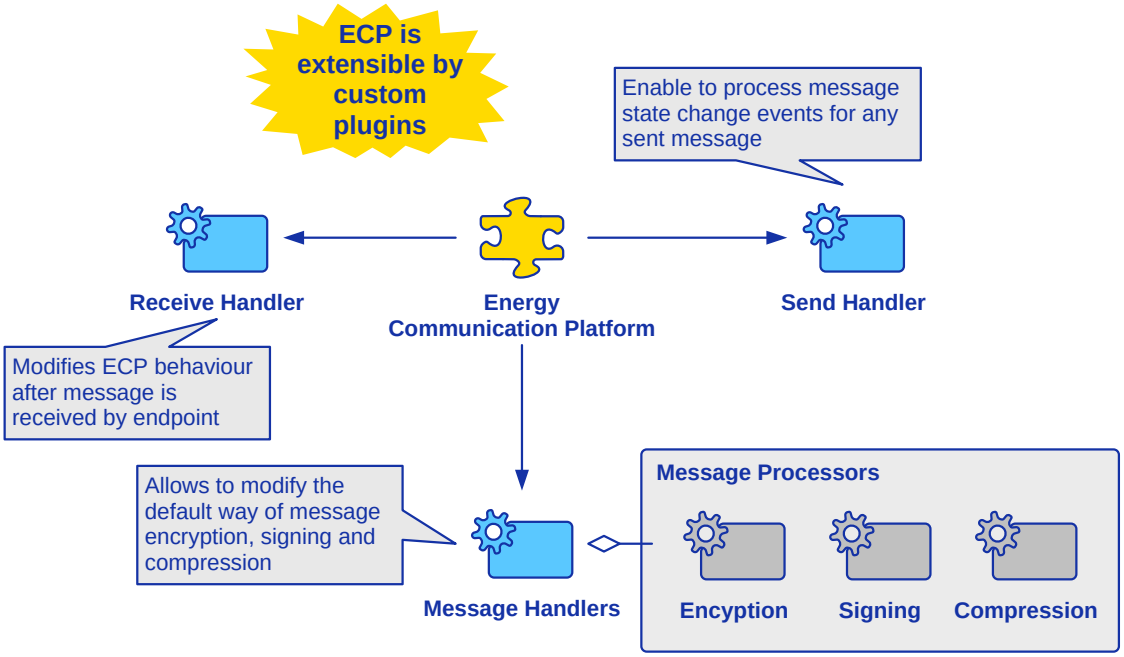

Figure 5: Extensibility

By writing custom plugins, one can influence how the ECP informs the business application about sent and received messages (Send and Receive handlers) and also modify the way of processing the message before sending and after receiving (Message handlers).

For example, by providing custom Receive handler, the ECP may directly resend the message to the web services of target business application right after the message is received by Endpoint.

Another example of extension could be providing a special Signing message processor that signs the message not using integrated certificate store, but instead uses connected chip card.

4.5. Overview of ECP Endpoint Interfacing

4.5.1. ECP Communication Interface

ECP Endpoint provides carefully designed API for integration with business applications that can then use ECP for sending and receiving messages.

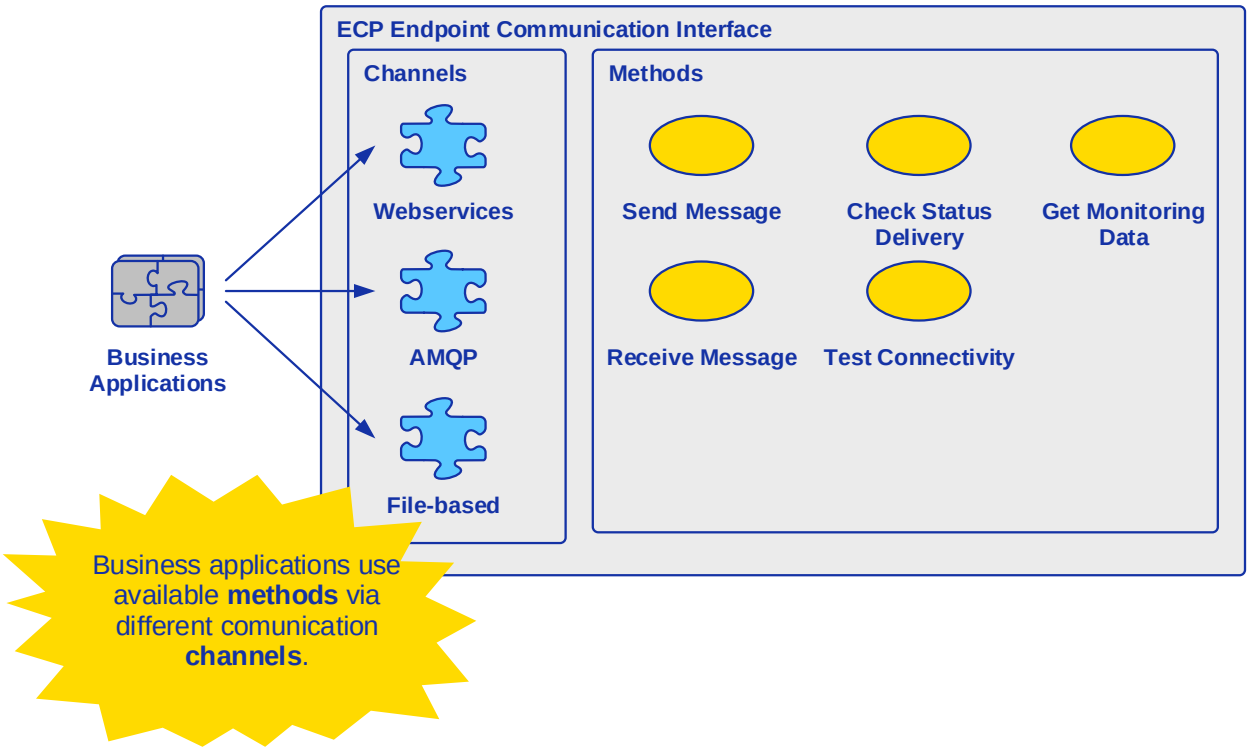

Figure 6: Endpoint API

The Endpoint API defines five methods:

- **Send message**, enabling business application to construct and send a message.
- **Receive message**, allowing the business application to receive a message and find out the number of waiting messages to be delivered.
- **Check delivery status**, allowing the sender to check the state of the message delivery – if it has been already delivered, if it is still on the way etc.
- **Test connectivit**y to find out whether some specific endpoint is reachable via ECP.
- **Get monitoring data** to gather data with information about status of the Endpoint – the applications may use it to verify if the Endpoint itself is in healthy state.

These methods are available through three different technological channels:

- **Webservices** – preferred way of integration, compatible with most of available technologies.
- **AMQP** – high performant API for sending and receiving messages. Only Send and Receive message methods are available via this channel.
- **File-based interface** – possible way of integration for applications that for any reason cannot use previous two channels. Only Send and Receive message methods are available via this channel. Message state is viewable via records in a log file.

The integration API is in depth documented in the ECP Public Interface document.

4.5.2. ECP User Interface

ECP Endpoint provides lightweight web-based GUI that can be used to access most ECP message-related features:

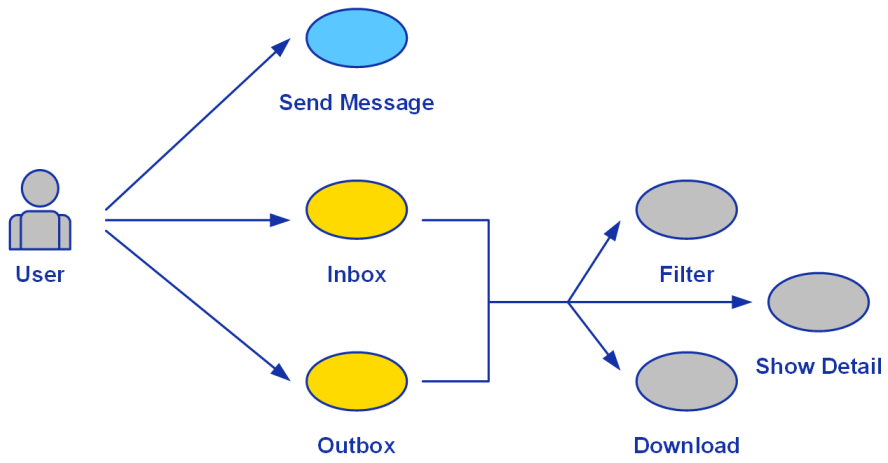

Figure 7: Endpoint GUI

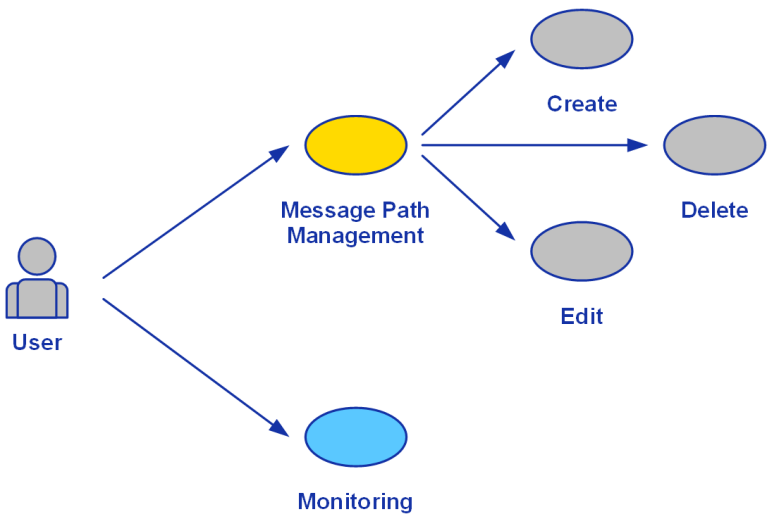

Figure 8: Endpoint GUI 2

The GUI functionalities provide the abstraction of Inbox and Outbox well-known from common mailing clients. It is also possible to compose and send message and read all received and sent messages. In addition, the user can Manage Message Paths and execute the Monitoring screen to check the status of ECP endpoint.

Message Path management enables user to define how certain messages will be transported via ECP network. It is possible to define Message Path specific for receiver and message type.

4.5.3. Communication Schema

ECP can send messages using indirect or direct communication pattern. The communication pattern is defined in message path.

4.5.3.1. Indirect

When sending and receiving messages using indirect communication pattern, ECP components communicate in following way:

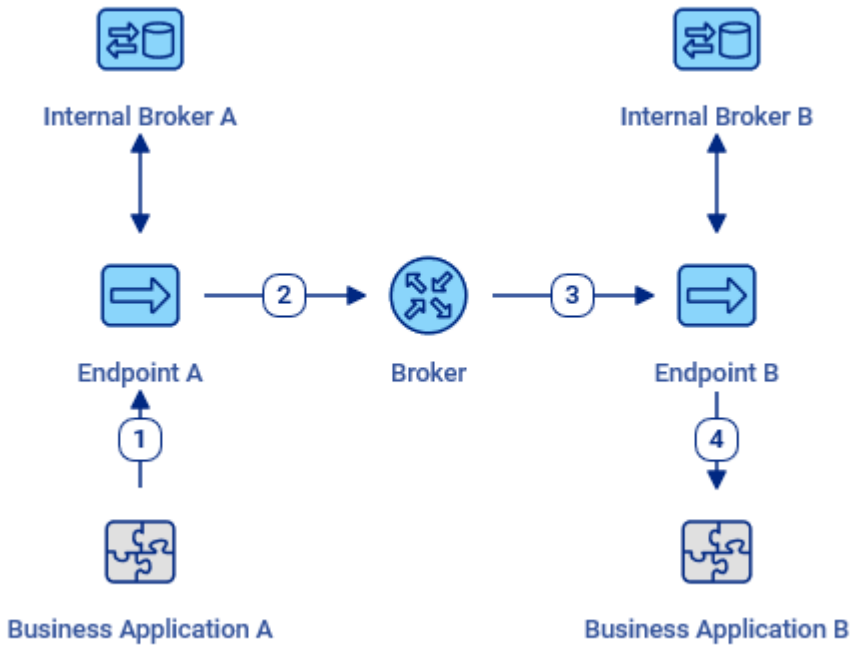

Figure 9: Indirect Communication

1. Business Application submits message into the ECP using the ECP Public API "Send message" method.
2. The ECP Endpoint encrypts the message with public encryption certificate of message recipient, signs it with its own private certificate and sends it to recipient via Broker configured in message path.

3. The message is passed to the recipient ECP Endpoint, which verifies the signature and decrypts it.
4. The message is passed to the target business application, which is the recipient of the communication.

4.5.3.2. Direct

When sending and receiving messages using direct communication pattern, ECP components communicate in following way:

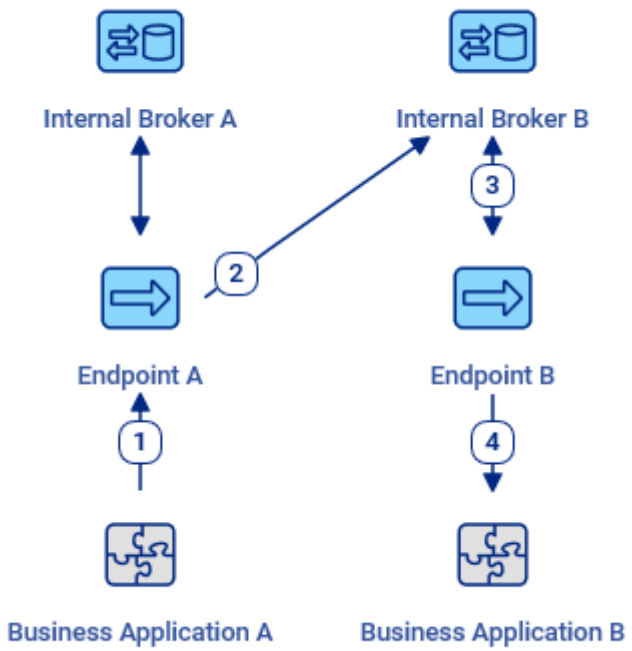

Figure 10: Direct Communication

1. Business Application submits message into the ECP using the ECP Public API "Send message" method.
2. The ECP Endpoint encrypts the message with public encryption certificate of message recipient, signs it with its own private certificate and sends it to the Internal Broker of the recipient Endpoint.
3. Recipient Endpoint receives message, verifies the signature, and decrypts it.
4. The message is passed to the target business application, which is the recipient of the communication.

After any Endpoint receives the message, an acknowledgement is sent back to sending Endpoint. This way, the sending Endpoint keeps track of any sent message. For sending acknowledgements ECP always use same message path as for original message.

4.6. Security Features

4.6.1. Main Concepts

ECP is designed to play role of highly reliable and secure message transfer system:

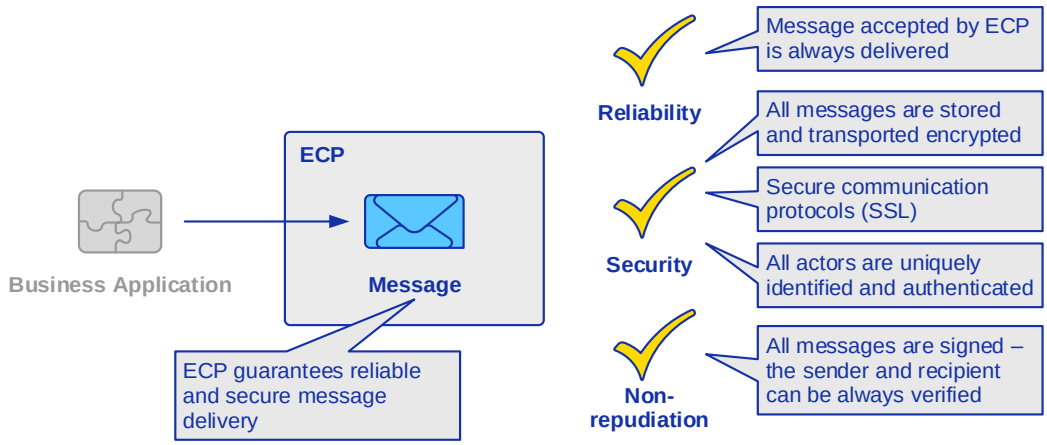

Figure 11: Security and Reliability

ECP guarantees that any message that was accepted by ECP will be delivered if the recipient is available. The sender can at any time verify the state of message delivery to find out if the message is still on the way (and where), or if it was already delivered. An acknowledgement is sent to the sender when the message is successfully delivered.

All messages are transported encrypted and signed. Information about message traversal is logged on all ECP message handling components; ECP provides non-repudiation feature enabling to verify the message and all its metadata including sender, recipient, times of sending, delivery etc.

On communication layer, ECP components use secure communication protocol (SSL) – any information is transported encrypted. Moreover, both sides of communication are always identified and authenticated by industry-standard PKI certificates.

For more information about security features, please see chapter *Security Features*.

4.6.2. Overview

Main goals of the ECP security can be summarized by the following points.

- The security solution is transparent to the business application – no specific implementation is required in the applications to communicate securely.
- Any message sent via the ECP is readable only by its recipient.

- The sender of any message can be unambiguously identified.
- Non-repudiation of the messages – it can be unambiguously proven that the sender sent the message and recipient received it
- All potentially unsecured communication routes are encrypted.
- The security solution is compatible with the X.509 public key infrastructure.

The security issues are covered on two levels: Transport-layer Security and Message Level Security.

On the transport layer, the ECP platform ensures that the communication between two ECP communication components is always encrypted, and both participants of the communication are unambiguously identified.

On the message level, the ECP provides message signing and encryption, so that the sender can be unambiguously identified and the message itself is readable only by its recipient.

All security features are described in detail in the System Design document, which also covers advanced issues, such as the certificate issuing process or certificate revocations.

4.6.3. Transport-layer Security

The security of the communication between ECP messaging components is ensured on the transport protocol layer. When communicating via the network, a secure protocol (HTTPS/SSL) is used, providing encryption of the communication route. Mutual authentication of communicating components is handled using X.509 certificates; both the client and the server authenticate themselves by their client or server X.509 certificates.

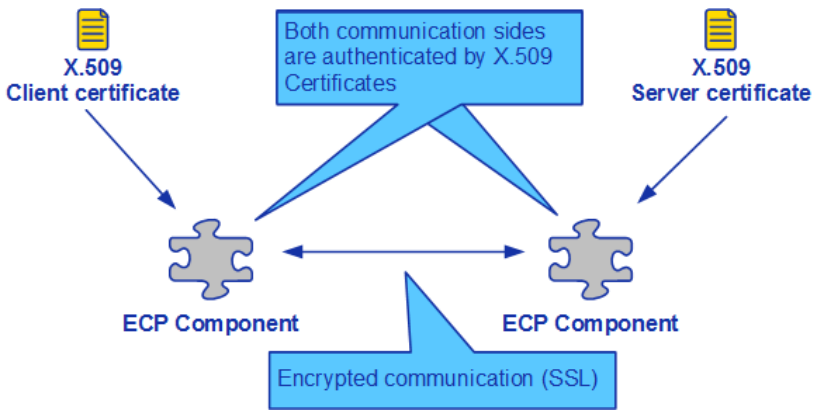

Figure 12: Transport Level Security

The communication between ECP messaging components is always initiated by the client, which provides higher level of security on client-side firewalls by not having to allow any incoming requests.

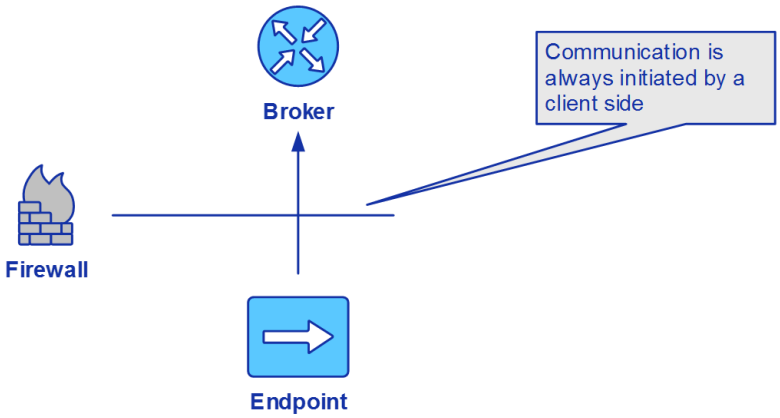

Figure 13: Security - firewall

However, if some Endpoint requires direct communication pattern, then the Internal Broker of this Endpoint must accept incoming request for specified ports and protocols.

4.6.4. Message-level Security

The unambiguous identification of the sender of any message sent via the ECP is enabled by usage of digital signatures:

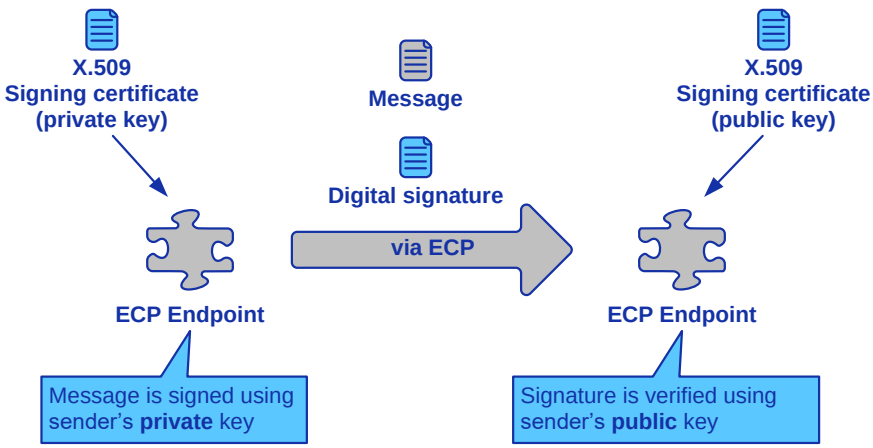

Figure 14: Message Security - signature

On the sender's Endpoint, the message is signed using the sender's private certificate. On the recipient's Endpoint, the message signature is verified using the sender's public key.

Any message sent via the ECP is encrypted, so that it is only its recipient that can read the message contents:

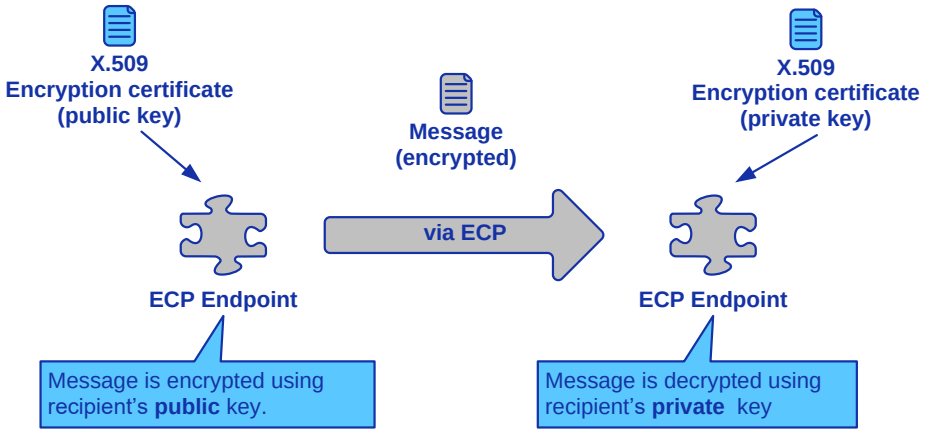

Figure 15: Message Security - encryption

The sender encrypts the message using the public encryption key of the recipient. This way, it is only the recipient who can decrypt the message (nobody else knows the private key).

In fact, the message is not encrypted with the public key directly, but with a generated session key, which is then encrypted with the public key and sent together with the message to the recipient. More details are provided in the System Design document.

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